

Umaa Maheswaran V

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About

AI engineer interested in building intelligent systems that go beyond the notebook. My experience spans deep learning, computer vision, agentic AI, and full stack AI applications using RAG, LangGraph, and MCP. I enjoy taking ideas from prototype to production and building reliable AI products with strong startup mindset.

Experience

Sanlayan Technologies (Defence Technology Startup)

Bangalore, India

AI/ML Intern

Mar 2026 – Present

- Generated synthetic multi-channel IQ radar data with configurable target, clutter, and noise parameters to build training datasets for a machine-learning-based radar detection model.
- Trained and evaluated multiple candidate ML detection models against conventional CFAR baselines, selecting the best-performing model based on detection accuracy across varying target and clutter conditions.
- Contributed to supporting radar signal-processing workflows, including feature extraction and real-time embedded interface development.

Technologies

Languages: Python, C/C++, SQL

AI/ML & Deep Learning: PyTorch, CNNs, GANs, OpenCV, Scikit-learn, NumPy, Pandas

LLM & Agentic AI: LangChain, LangGraph, Model Context Protocol (MCP), Context Engineering

Systems & Deployment: CUDA, Docker, Linux, Git

Projects

Memora — AI-Powered Flashcard Learning Platform

Tools: FastAPI, React, PostgreSQL (Neon), MCP, Groq API, Render, Vercel | [Live Demo](#)

- Built and deployed a full-stack flashcard platform using FastAPI, React, and Neon PostgreSQL, implementing a spaced-repetition algorithm for adaptive review scheduling across user-created flashcards.
- Integrated Model Context Protocol (MCP) with an LLM-powered flashcard generation pipeline, enabling AI assistants to create, review, and manage flashcards conversationally from raw study notes.

Enterprise Knowledge Agent (RAG)

Tools: Python, LangGraph, FAISS, Google Gemini Flash, Gradio, Hugging Face Spaces

- Developed a retrieval-augmented generation (RAG) agent using LangGraph and FAISS to answer enterprise knowledge queries over company documentation with grounded, context-aware responses.
- Designed a multi-stage workflow with semantic retrieval, LLM response generation, grounding validation, and an evaluation framework using automated and LLM-based grading, then deployed the application.

Edge-Optimized Generative Image Model

Tools: Python, PyTorch, OpenCV, GANs

- Designed a lightweight generative image model optimized for edge deployment using structured pruning and post-training quantization.
- Built efficient image preprocessing and inference workflows for low-latency computer vision applications.

Education

Amrita Vishwa Vidyapeetham

B.Tech in Electronics and Communication Engineering

CGPA: 7.63/10 (8 Semesters)

2022 – 2026

CS Academy

Higher Secondary Education

Class 12: 86.2% Class 10: 87.6%

2019 – 2022

Certifications

AI on Jetson Nano — NVIDIA

Supervised Machine Learning — Stanford University and DeepLearning.AI

Achievements

Hackathon: Finalist — ELCIA All India Sense2Scale Hackathon (Top 10 out of 200+ teams across India).

Speed Cubing: Competitive speed cuber– can solve a Rubik's Cube in under 15 seconds.